Project Folder Structure

```
mern-auth-project
│
├── backend
│   ├── server.js
│   ├── models
│   │   User.js
│   ├── routes
│   │   auth.js
│   ├── middleware
│   │   auth.js
│
└── frontend
    └── src
        ├── App.js
        ├── Login.js
        ├── Register.js
```

Backend Setup

Install Packages

Inside **backend folder**

```
npm init -y
npm install express mongoose cors bcryptjs jsonwebtoken
```

---

server.js

```
const express = require("express")
const mongoose = require("mongoose")
const cors = require("cors")

const app = express()

app.use(express.json())
app.use(cors())

mongoose.connect("mongodb://127.0.0.1:27017/authdb")
.then(()=>console.log("MongoDB Connected"))

app.use("/api",require("./routes/auth"))

app.listen(5000,()=>console.log("Server running on port 5000"))
```

User Model

backend/models/User.js

```
const mongoose = require("mongoose")

const UserSchema = new mongoose.Schema({
 name:String,
 email:String,
 password:String
})

module.exports = mongoose.model("User",UserSchema)
```

JWT Middleware

backend/middleware/auth.js

```
const jwt = require("jsonwebtoken")

module.exports = function(req,res,next){

 const token = req.header("Authorization")

 if(!token){
  return res.status(401).json({msg:"No token"})
 }

 try{

  const decoded = jwt.verify(token,"secret123")

  req.user = decoded

  next()

 }catch(err){
  res.status(401).json({msg:"Invalid token"})
 }

}
```

Authentication Routes

backend/routes/auth.js

```
const express = require("express")
const router = express.Router()
```

```
const bcrypt = require("bcryptjs")
const jwt = require("jsonwebtoken")

const User = require("../models/User")
const auth = require("../middleware/auth")

// Register
router.post("/register",async(req,res)=>{

 const {name,email,password} = req.body

 const hashed = await bcrypt.hash(password,10)

 const user = new User({
  name,
  email,
  password:hashed
 })

 await user.save()

 res.json({msg:"User Registered"})

})


// Login
router.post("/login",async(req,res)=>{

 const {email,password} = req.body

 const user = await User.findOne({email})

 if(!user){
  return res.status(400).json({msg:"User not found"})
 }

 const isMatch = await bcrypt.compare(password,user.password)

 if(!isMatch){
  return res.status(400).json({msg:"Wrong password"})
 }

 const token = jwt.sign(
  {id:user._id},
  "secret123",
  {expiresIn:"1h"}
 )

 res.json({token})

})


// Protected Route
router.get("/profile",auth,async(req,res)=>{
```

```
 const user = await User.findById(req.user.id).select("-password")

 res.json(user)

})

module.exports = router
```

Frontend Setup

Create React project

```
npx create-react-app frontend
cd frontend
npm install axios
```

App.js

frontend/src/App.js

```
import React from "react"
import Login from "./Login"
import Register from "./Register"

function App(){
 return(
  <div>
   <h1>MERN Authentication</h1>
   <Register/>
   <Login/>
  </div>
 )
}

export default App
```

Register Component

frontend/src/Register.js

```
import React,{useState} from "react"
import axios from "axios"

function Register(){

 const [name,setName] = useState("")
 const [email,setEmail] = useState("")
```

```
 const [password,setPassword] = useState("")

 const register = async()=>{

  await axios.post("http://localhost:5000/api/register",{
   name,
   email,
   password
  })

  alert("User Registered")

 }

 return(

  <div>

   <h2>Register</h2>

   <input placeholder="name"
   onChange={e=>setName(e.target.value)} />

   <input placeholder="email"
   onChange={e=>setEmail(e.target.value)} />

   <input placeholder="password"
   type="password"
   onChange={e=>setPassword(e.target.value)} />

   <button onClick={register}>Register</button>

  </div>

 )

}

export default Register
```

Login Component

frontend/src/Login.js

```
import React,{useState} from "react"
import axios from "axios"

function Login(){

 const [email,setEmail] = useState("")
 const [password,setPassword] = useState("")

 const login = async()=>{
```

```
  const res = await axios.post(
   "http://localhost:5000/api/login",
   {email,password}
  )

  localStorage.setItem("token",res.data.token)

  alert("Login Successful")

 }

 return(

  <div>

   <h2>Login</h2>

   <input placeholder="email"
   onChange={e=>setEmail(e.target.value)} />

   <input placeholder="password"
   type="password"
   onChange={e=>setPassword(e.target.value)} />

   <button onClick={login}>Login</button>

  </div>

 )

}

export default Login
```

Access Protected Route

Example request

```
const token = localStorage.getItem("token")

axios.get("http://localhost:5000/api/profile",{
 headers:{
  Authorization: token
 }
}).then(res=>{
 console.log(res.data)
})
```

Response example

```
{
 name: "John",
```

```
 email: "john@mail.com"
}
```

MongoDB Example Document

```
{
 "_id":"66578abc",
 "name":"John",
 "email":"john@mail.com",
 "password":"$2a$10$ksjdhkjsd..."
}
```

Password is **hashed using bcrypt**.

How To Run Project

Start MongoDB

```
mongod
```

Run Backend

```
cd backend
node server.js
```

Server

http://localhost:5000

Run React

```
cd frontend
npm start
```

React

http://localhost:3000